COMPETITIVE PROGRAMMING

1-Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

```
#include <stdio.h>

int main() {
    int n, i, j;
    scanf("%d", &n);

    int a[n];

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (a[i] == a[j]) {
                printf("%d", a[i]);
                return 0;
            }
        }
    }

    return 0;
}
```

2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

```
#include <stdio.h>
int main() {
```

```
    int n, i, x;
    int sum = 0;
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        scanf("%d", &x);
        sum += x;
    }
    printf("%d", sum - (n * (n - 1)) / 2);
    return 0;
}
```

3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity

```
#include <stdio.h>
int main()
{
    int n, m, a[100], b[100];
    scanf("%d", &n);
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &m);
    for (int i = 0; i < m; i++)
        scanf("%d", &b[i]);
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < m; j++)
        {
            if (a[i] == b[j])
            {
```

```
                printf("%d ", a[i]);
                break;
            }
        }
    }
    return 0;
}
```

4-Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Complexity

```
#include <stdio.h>
int main() {
    int t, n, m, i, j;
    scanf("%d", &t);
    while (t--) {
        scanf("%d", &n);
        int a[n];
        for (i = 0; i < n; i++)
            scanf("%d", &a[i]);
        scanf("%d", &m);
        int b[m];
        for (i = 0; i < m; i++)
            scanf("%d", &b[i]);
        i = 0;
        j = 0;
        while (i < n && j < m) {
            if (a[i] == b[j]) {
                printf("%d ", a[i]);
```

```
            i++;
            j++;
        }
        else if (a[i] < b[j])
            i++;
        else
            j++;
    }
    printf("\n");
  }
  return 0;
}
```

5-Pair with Difference-O(n^2)Time Complexity,O(1) Space Complexity

```
#include <stdio.h>
int main() {
  int n, i, j, k;
  scanf("%d", &n);
  int a[n];
  for (i = 0; i < n; i++)
    scanf("%d", &a[i]);
  scanf("%d", &k);
  for (i = 0; i < n; i++) {
    for (j = i + 1; j < n; j++) {
      if (a[j] - a[i] == k || a[i] - a[j] == k) {
        printf("1");
        return 0;
      }
```

```
        }
    }
    printf("0");
    return 0;
}
```

6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

```
#include <stdio.h>
int main() {
    int n, k;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &k);
    int i = 0, j = 1;
    while (j < n) {
        int diff = a[j] - a[i];
        if (diff == k) {
            printf("1");
            return 0;
        } else if (diff < k) {
            j++;
        } else {
            i++;
            if (i == j)
                j++;
```

```
        }
    }
    printf("0");
    return 0;
}
```